

COVARIANCE, VARIANCE OF SUMS, AND CORRELATION

QUESTION

Given X, Y : rand vars in a prob space, what is $\text{Var}(X+Y)$?

Solution

$$\begin{aligned} \text{By def, } \text{Var}(X+Y) &= \mathbb{E}[(X+Y)^2] - \mathbb{E}[X+Y]^2 \stackrel{\text{lin of exp}}{=} \mathbb{E}[X^2 + 2XY + Y^2] - (\mathbb{E}[X] + \mathbb{E}[Y])^2 \stackrel{\text{lin of exp}}{=} \mathbb{E}[X^2] + 2\mathbb{E}[XY] + \mathbb{E}[Y^2] - \mathbb{E}[X]^2 - 2\mathbb{E}[X]\mathbb{E}[Y] - \mathbb{E}[Y]^2 \\ &= (\mathbb{E}[X^2] - \mathbb{E}[X]^2) + (\mathbb{E}[Y^2] - \mathbb{E}[Y]^2) + 2(\mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]) \\ &\stackrel{\text{by def}}{=} \text{Var}(X) + \text{Var}(Y) + 2(\mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]) \end{aligned}$$

PROPOSITION

Let X, Y be indep and $g, h: \mathbb{R} \rightarrow \mathbb{R}$ are arbitrary.

Then, $\mathbb{E}[g(X)h(Y)] = \mathbb{E}[g(X)]\mathbb{E}[h(Y)]$

Proof for conti case

$$\text{By def, } \mathbb{E}[g(X)h(Y)] = \int_{\mathbb{R}^2} g(x)h(y) f_{X,Y}(x,y) dx dy$$

$$\begin{aligned} \text{By indep, } &= \int_{\mathbb{R}^2} g(x)h(y) \underline{f_X(x)f_Y(y)} dx dy \\ &= \int_{\mathbb{R}} g(x)f_X(x) dx \int_{\mathbb{R}} h(y)f_Y(y) dy \end{aligned}$$

$$\text{By def, } = \mathbb{E}[g(X)]\mathbb{E}[h(Y)] \quad \square$$

COROLLARY

If X, Y : indep, then $\text{Var}(X+Y) = \text{Var}(X) + \text{Var}(Y)$

Proof

As X, Y indep, by prop, $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$

$$\therefore \text{Var}(X+Y) = \text{Var}(X) + \text{Var}(Y) + 2(\mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]) \stackrel{\rightarrow 0}{=} \text{Var}(X) + \text{Var}(Y) \quad \square$$

EXAMPLE

For $X \sim \text{Bin}(n, p)$, $X = X_1 + X_2 + \dots + X_n$, where $X_i \sim \text{Ber}(p)$: indep

As $\text{Var}(X_i) = p(1-p) \quad \forall i$, thus $\text{Var}(X) = \sum_{i=1}^n \text{Var}(X_i) = \sum_{i=1}^n p(1-p) = np(1-p) \quad \checkmark$

EXAMPLE

For indep $X \sim N(\mu_1, \sigma_1^2)$, $Y \sim N(\mu_2, \sigma_2^2)$, then $X+Y \sim N(\mu_1+\mu_2, \sigma_1^2+\sigma_2^2)$

Notice, $\text{Var}(X+Y) = \sigma_1^2 + \sigma_2^2 = \text{Var}(X) + \text{Var}(Y)$

DEFINITION

Given rand vars X, Y in a prob space, the covariance of X and Y is defined as $\text{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])]$, recall: $\text{Var}(X) := \mathbb{E}[(X - \mathbb{E}[X])^2]$

OBSERVE

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])]$$

$$= \mathbb{E}[XY - X\mathbb{E}[Y] - Y\mathbb{E}[X] + \mathbb{E}[X]\mathbb{E}[Y]]$$

$$= \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] + \mathbb{E}[X]\mathbb{E}[Y] + \mathbb{E}[X]\mathbb{E}[Y] \quad (\text{lin of exp})$$

$$= \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$$

OBSERVE

$$\textcircled{1} \text{Var}(X+Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y)$$

$$\textcircled{2} \text{ If } X, Y: \text{ indep, then } \text{Cov}(X, Y) = 0$$

COUNTEREXAMPLE (Cov=0 $\nRightarrow$ INDEP)

$$\text{Let } X = \begin{cases} -1, & \text{w.p. } \frac{1}{3} \\ 0, & \text{w.p. } \frac{2}{3} \end{cases} \quad Y = \begin{cases} 1, & X=0 \\ 0, & X \neq 0 \end{cases}$$

Notice, $XY \equiv 0$, so $\mathbb{E}[XY] = 0$. Also, $\mathbb{E}[X] = (-1)(\frac{1}{3}) + 0(\frac{2}{3}) = -\frac{1}{3}$, $\mathbb{E}[Y] = 0 \Rightarrow \mathbb{E}[X]\mathbb{E}[Y] = 0 \therefore \text{Cov}(X, Y) = 0$

Claim: X, Y : NOT indep

Proof

$$P(X=1, Y=1) = 0 \text{ since } X=1 \Rightarrow X \neq 0 \Rightarrow Y=0$$

$$\text{However, } P(X=1) = \frac{1}{2} \text{ by def, } P(Y=1) = P(X=0) = \frac{1}{2}$$

$$\therefore P(X=1, Y=1) \neq P(X=1)P(Y=1) \Rightarrow \text{not indep} \quad \square$$

PROPERTIES OF COVARIANCE

Let $X, Y, X_1, \dots, X_n, Y_1, \dots, Y_m$ be rand var

$$(i) \text{ Cov}(X, Y) = \text{Cov}(Y, X)$$

$$(ii) \text{ Cov}(X, X) = \text{Var}(X)$$

$$(iii) \text{ Cov}(aX, Y) = a \text{Cov}(X, Y) \quad \forall a \in \mathbb{R}$$

$$(iv) \text{ Cov}\left(\sum_{i=1}^n X_i, \sum_{j=1}^m Y_j\right) = \sum_{i=1}^n \sum_{j=1}^m \text{Cov}(X_i, Y_j)$$

Proof

(i) - (iii) are immediate by def, so we will just formally prove (iv).

$$\text{By def, } \text{Cov}\left(\sum_{i=1}^n X_i, \sum_{j=1}^m Y_j\right) = E\left[\left(\sum_{i=1}^n X_i\right)\left(\sum_{j=1}^m Y_j\right)\right] - E\left[\sum_{i=1}^n X_i\right]E\left[\sum_{j=1}^m Y_j\right]$$

$$\text{Fubini?} \Rightarrow E\left[\sum_{i=1}^n \sum_{j=1}^m X_i Y_j\right] - E\left[\sum_{i=1}^n X_i\right]E\left[\sum_{j=1}^m Y_j\right]$$

$$\text{lin of exp} \Rightarrow \sum_{i=1}^n \sum_{j=1}^m E[X_i Y_j] - \left(\sum_{i=1}^n E[X_i]\right)\left(\sum_{j=1}^m E[Y_j]\right)$$

$$= \sum_{i=1}^n \sum_{j=1}^m (E[X_i Y_j] - E[X_i]E[Y_j])$$

$$= \sum_{i=1}^n \sum_{j=1}^m \text{Cov}(X_i, Y_j) \quad \square$$

COROLLARY

$$\text{If } X = \sum_{i=1}^n X_i, \text{ then } \text{Var}(X) = \sum_{i=1}^n \text{Var}(X_i) + 2 \sum_{i < j} \text{Cov}(X_i, X_j)$$

Proof

$$\text{By prop, } \text{Var}(X) = \text{Cov}(X, X) = \text{Cov}\left(\sum_{i=1}^n X_i, \sum_{j=1}^n X_j\right) = \sum_{i=1}^n \sum_{j=1}^n \text{Cov}(X_i, X_j) = \sum_{i=1}^n \text{Cov}(X_i, X_i) + \sum_{i \neq j} \text{Cov}(X_i, X_j) = \sum_{i=1}^n \text{Var}(X_i) + 2 \sum_{i < j} \text{Cov}(X_i, X_j) \quad \square$$

COROLLARY

If $X_1, \dots, X_n$ are pairwise indep, then $\text{Var}\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n \text{Var}(X_i)$ no need for mutual indep

Proof

It follows immediately from pairwise indep $\Rightarrow \text{Cov}(X_i, X_j) = 0 \quad \forall i \neq j$

EXAMPLE

We collect independent samples $X_1, \dots, X_n$ from some distribution X with mean $= \mu$ and variance $= \sigma^2$

We define $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$ to be the sample mean and $S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$ to be the sample variance

$$\text{Then, } E[\bar{X}] = \frac{1}{n} \sum_{i=1}^n E[X_i] = \frac{1}{n} \sum_{i=1}^n \mu = \mu$$

↑ lin of exp

We say $\bar{X}$ is an unbiased estimator of μ because it has the right value in expectation

$$\text{Notice, } \text{Var}(\bar{X}) = \frac{1}{n^2} \text{Var}\left(\sum_{i=1}^n X_i\right) = \frac{1}{n^2} \sum_{i=1}^n \text{Var}(X_i) = \frac{1}{n^2} \sum_{i=1}^n \sigma^2 = \frac{\sigma^2}{n}, \text{ which decreases as } n \text{ increases!} \rightarrow \text{why we use repeated trials}$$

↑ indep

$$\text{Here, } E[(n-1)S^2] = E\left[\sum_{i=1}^n (X_i - \bar{X})^2\right] = E\left[\sum_{i=1}^n (X_i^2 - 2X_i \bar{X} + \bar{X}^2)\right] = E\left[\sum_{i=1}^n X_i^2 - 2\bar{X} \sum_{i=1}^n X_i + n\bar{X}^2\right] = E\left[\sum_{i=1}^n X_i^2 - n\bar{X}^2\right]$$

$$\text{By lin of exp, } = \sum_{i=1}^n E[X_i^2] - nE[\bar{X}^2]$$

$$\text{By def of Var, } = \sum_{i=1}^n (\text{Var}(X_i) + E[X_i]^2) - n(\text{Var}(\bar{X}) + E[\bar{X}]^2)$$

$$= n(\sigma^2 + \mu^2) - n\left(\frac{\sigma^2}{n} + \mu^2\right)$$

$$= (n-1)\sigma^2$$

$$\therefore E[S^2] = \sigma^2, \text{ i.e. } S^2 \text{ is an unbiased estimator of } \sigma^2.$$

DEFINITION

Given rand var X, Y on a prob space, with $\sigma_X = \sqrt{\text{Var}(X)}$ and $\sigma_Y = \sqrt{\text{Var}(Y)}$, we define the correlation of X and Y as $\rho(X, Y) = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X) \text{Var}(Y)}}$

PROPOSITION

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For any random vars X, Y , $-1 \leq \rho(X, Y) \leq 1$

Proof

Consider $\text{Var}(\frac{X}{\sigma_X} + \frac{Y}{\sigma_Y})$

Then, $\text{Var}(\frac{X}{\sigma_X} + \frac{Y}{\sigma_Y}) = \text{Var}(\frac{X}{\sigma_X}) + \text{Var}(\frac{Y}{\sigma_Y}) + 2\text{Cov}(\frac{X}{\sigma_X}, \frac{Y}{\sigma_Y}) = \frac{1}{\sigma_X^2} \text{Var}(X) + \frac{1}{\sigma_Y^2} \text{Var}(Y) + \frac{2\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$

By def, $= 2 + 2\rho(X, Y)$

As $\text{Var}(\cdot) \geq 0$, thus $\text{Var}(\frac{X}{\sigma_X} + \frac{Y}{\sigma_Y}) = 2 + 2\rho(X, Y) \geq 0 \Rightarrow \rho(X, Y) \geq -1$

Now, consider $\text{Var}(\frac{X}{\sigma_X} - \frac{Y}{\sigma_Y})$,

Similarly, $0 \leq \text{Var}(\frac{X}{\sigma_X} - \frac{Y}{\sigma_Y}) = 2 - 2\rho(X, Y) \Rightarrow \rho(X, Y) \leq 1 \quad \square$

EQUALITY CASES

If $\rho(X, Y) = 1$, thus $2 - 2\rho(X, Y) = \text{Var}(\frac{X}{\sigma_X} - \frac{Y}{\sigma_Y}) = 0$

$\therefore \frac{X}{\sigma_X} - \frac{Y}{\sigma_Y} = c$, c : const

$\therefore Y = \frac{\sigma_Y}{\sigma_X} X - c\sigma_Y$, i.e. $Y = aX + b$ for some $a > 0$ ($\because \sigma_X, \sigma_Y > 0$)

Similarly, if $\rho(X, Y) = -1$, thus $2 + 2\rho(X, Y) = \text{Var}(\frac{X}{\sigma_X} + \frac{Y}{\sigma_Y}) = 0$

$\therefore Y = aX + b$ for some $a < 0$

RECALL

X, Y : indep $\Rightarrow \text{Cov}(X, Y) = 0 \Leftrightarrow \rho(X, Y) = 0$ but the converse is not true

DEFINITION

If $\rho(X, Y) = 0$, we say X and Y are unrelated (doesn't mean indep, just means no linear relationship)

EXAMPLE

Let A, B be events in a prob space, and let $\mathbb{1}_A, \mathbb{1}_B$ be their indicator rand vars

Then, $\text{Cov}(\mathbb{1}_A, \mathbb{1}_B) = \mathbb{E}[\mathbb{1}_A \mathbb{1}_B] - \mathbb{E}[\mathbb{1}_A] \mathbb{E}[\mathbb{1}_B]$

$$= \mathbb{E}[\mathbb{1}_{A \cap B}] - \mathbb{E}[\mathbb{1}_A] \mathbb{E}[\mathbb{1}_B]$$

$$= P(A \cap B) - P(A)P(B)$$

By def, $= P(B)[P(A|B) - P(A)]$